

# Giuliana Cabrera Sanchez

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## EDUCATION

### Massachusetts Institute of Technology (MIT) | Cambridge, MA

Candidate for B.S. in Computer Science and Electrical Engineering

Class of May 2025

*Relevant Courses:* Digital Systems, Microcomputer Project Laboratory (Circuit Design, Chips, Microcontrollers),

Computation Structures (RISC-V, Minispec), Software Construction, Algorithms, Digital Systems (FPGAs)

MIT MISTI Travel Fellowship for Research | UK & Brazil

Sept to May 2026

## EXPERIENCE

### Research Intern, Queen's University Belfast | Belfast, UK

Sept 2025 - Dec 2025

- Integrated URJA, a shared-library tool for dynamic voltage and frequency scaling and hardware performance counter collection, as a SPANK plug-in for the Slurm job scheduler, enabling node-level energy profiling and runtime CPU frequency control.
- Developed Linux shell tooling and automation scripts to deploy, configure, and validate power-management behavior across distributed compute nodes under real workloads.

### Software Development Intern, Qualcomm | San Diego, CA

June 2025 - Aug 2025

- Built a hardware-aware profiling framework for Snapdragon SoCs using on-chip performance counters to measure power consumption, cache behavior, and CPU/GPU execution stalls.
- Designed automated pipelines in Jupyter to collect, parse, and analyze low-level telemetry from embedded processors, supporting evaluation and tuning of DVFS and clock-management algorithms.

### Undergraduate Researcher, MIT Global Engineering and Research Lab | Cambridge, MA

Feb 2025 - May 2025

- Designed and programmed hardware drivers in Python for sensors and actuators within an off-grid, containerized electrodialysis desalination system.
- Implemented and debugged communication protocols, including RS-485 and MQTT, to enable a reliable supervisory control of the pumps, valves, and sensors system with automated testing scripts.

### Software Engineering Intern, Qualcomm | San Diego, CA

June 2024 - Aug 2024

- Developed a new raytracing feature through bounding volume hierarchies as part of the graphics tools team for Qualcomm's Snapdragon Profiler.

### Software Engineering Intern, Slack (Salesforce) | San Francisco, CA

June 2023 - Aug 2023

- Enhanced Slack's calling function by introducing a native menu feature for macOS in Typescript and Redux, building it on the Electron framework.

## PROJECTS

### MITOSIS Sensor Node (Firmware Lead)

May 2025

- Led firmware design for a compact solar-powered temperature and humidity sensing device using ESP32-C3, e-ink display, and LoRa mesh networking.
- Implemented fault/error detection, deep sleep scheduling, and RTC-based data archiving (~15-day buffer).
- Integrated sensor drivers (SHT40, VEML7700) and display refresh logic in C++, optimizing for low power and modular expansion.

### PSoC 5LP Piano Tiles

May 2024

- Developed an interactive Piano Tiles game using a PSoC microcontroller, integrating a TFT display for gameplay.
- Implemented game logic, input handling, and timing-critical display updates in C and Assembly, including interrupt-driven button debouncing and efficient partial screen redraws.

### Smart Dungeons and Dragons Game

May 2022

- Designed embedded gameplay logic across ESP32 devices, implementing state-machine-driven player actions such as dice rolling, combat mechanics, and defensive interactions.

## LEADERSHIP / VOLUNTEERING

Vice President, MIT Latino Cultural Center

Sept 2022 - May 2025

Secretary, MIT Society of Hispanic Professional Engineers

Aug 2021 - May 2025

Former Vice President of DEI and President, MIT Alpha Chi Omega

Dec 2022 - Dec 2024

## SKILLS

**Programming Languages:** Python, C/C++, Typescript, SystemVerilog, Minispec, Bash, Assembly Language

**Other:** Git, TeraTerm, ExpressPCB, kiCAD,

Ubuntu/Linux, Docker, SSH, UART, MQTT

## HONORS

Member of Rewriting the Code + ColorStack, The Jack Kent Cooke Foundation College Scholar, Hispanic Scholarship Fund Scholar